

Can LLMs See What I See? A Study on Five Prompt Engineering Techniques for Evaluating UX on a Shopping Site

Subin Shin
Department of Cognitive Science
Yonsei University
Seoul, Republic of Korea
tlstnqls16@yonsei.ac.kr

Jeesun Oh
Department of Industrial Design
KAIST
Daejeon, Republic of Korea
sun.oh@kaist.ac.kr

Sangwon Lee
Department of Integrated Design
Yonsei University
Seoul, Republic of Korea
sangwon.lee@yonsei.ac.kr

Abstract

Usability testing is essential for improving digital user experiences but has practical limitations in terms of cost-effectiveness. Recent advancements in multimodal Large Language Models (LLMs), like ChatGPT-4, offer new possibilities for UX evaluations. This study investigated the most effective prompt engineering techniques for identifying UX issues in digital interfaces. To achieve this, five prompt engineering techniques were carefully selected based on previous research, and the outputs generated using these techniques were analyzed based on severity assessment criteria. We discovered that Role Prompting and (Zero-Shot) Chain of Thought Prompting were highly effective. Further investigation revealed that a hybrid approach combining both techniques produced the best results. Our findings shed light on the possibility of using multimodal LLM as a UX evaluator, offering meaningful value for future advancements in LLM-based UX evaluations.

CCS Concepts

• Human-centered computing → Usability testing.

Keywords

UX Evaluation, Large Language Model (LLM), Multimodal LLM, Prompt Engineering, Role Prompting, (Zero-Shot) Chain of Thought Prompting

ACM Reference Format:

Subin Shin, Jeesun Oh, and Sangwon Lee. 2025. Can LLMs See What I See? A Study on Five Prompt Engineering Techniques for Evaluating UX on a Shopping Site. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720079>

1 Introduction

Usability testing is a widely used method among practitioners to enhance User Experiences (UX) with digital products [6]. This allows designers to identify issues that either enhance or hinder user experiences by observing how users interact with digital products or provide feedback [24]. While usability testing is a valuable method,

it often requires significant time and effort, particularly in recruiting participants and conducting tasks, which poses challenges for small and medium-sized enterprises (SMEs) to implement extensively [9]. Additionally, the process can be resource-intensive when examining large volumes of digital products or uncovering edge cases beyond the main user flow, making it challenging to conduct repeated tests and validate the effectiveness of solutions [8].

With the advancement of multimodal Large Language Model (LLM) technologies such as ChatGPT-4 and Gemini [1, 30], the potential to leverage integrated text-image datasets in LLM for precise UX evaluation is increasing [15, 18]. These models heavily rely on how input data is interpreted and processed, making the design and implementation of prompt engineering techniques essential [5, 26]. However, previous studies on utilizing LLM for UX evaluation have been underexplored to demonstrating their potential as collaborative tools or analyzing interview-based text data [7, 14]. Research on prompt engineering techniques for UX evaluation remains insufficient, particularly in developing effective strategies for multimodal environments that combine text and visual elements. Therefore, exploring effective prompt engineering techniques for UX evaluation using multimodal LLMs is crucial.

Our study aims to explore the effective prompt engineering techniques for UX evaluation using multimodal LLM, ChatGPT-4. The main findings of the study are as follows. Firstly, the Role Prompting technique effectively provided user-centered insights by assigning the LLM the perspective of a specific user persona. Then, (Zero-Shot) Chain of Thought (CoT) Prompting enabled systematic problem identification by step-by-step analysis of complex workflows. In addition, a hybrid approach combining Role Prompting and (Zero-Shot) CoT was particularly effective in identifying UX issues. Lastly, our findings revealed notable differences between human evaluators and LLM in UX assessment. While the LLM identified detailed issues that human evaluators often overlooked, it occasionally demonstrated hyper-accuracy distortions, overemphasizing minor issues or suggesting overly precise interpretations. Based on these findings, we discuss the challenges in UX evaluation that can be addressed using multimodal LLM-based tools and how these tools can help supplement existing processes. Our study contributes by offering valuable insights and proposing effective prompt engineering approaches that UX researchers and design practitioners can utilize effectively.

2 Related Works

2.1 Prompt Engineering

A prompt refers to input data that guides generative AI models to produce desired outputs, serving as a key element in effectively

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720079>

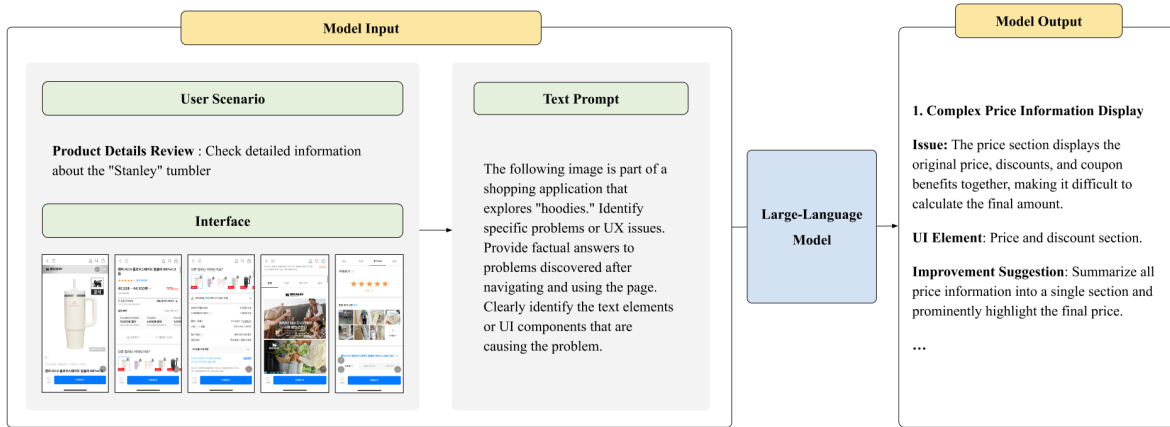


Figure 1: Multimodal Large Language Model UX Evaluation Process in Two Stages: (1) Presenting a user scenario and interface to identify specific UX issues; (2) Using a text prompt to guide the model in analyzing and providing improvement suggestions. The identified issues and improvement suggestions are based on the model's interpretation of the interface and user scenario.

directing the model's responses [22, 34]. Prompts can consist of text, images, sounds, or other media, and they play a crucial role in communicating user intent to ensure the model operates as intended [4]. Prompt engineering is the technique of designing and optimizing such prompts to utilize Language Models (LMs) efficiently across various applications and research domains [3]. With the technological advancements of LLMs like ChatGPT-4, the importance and interest in prompt engineering have grown significantly [11, 31]. Schulhoff et al. [28], through a systematic review, identified 58 text-based prompt engineering techniques (e.g., Few-Shot Prompting, Tree of Thought Prompting) and highlighted their ability to optimize generative AI model performance and their effectiveness across diverse application areas. In this study, we reviewed previous studies on prompt engineering techniques to identify those most applicable to the field of UX evaluation, focusing on the five key techniques listed below [13, 21, 32, 33, 36].

2.2 Five Prompt Engineering Techniques for UX Evaluations

2.2.1 Zero-Shot Prompting. A technique in which the LM performs tasks based on simple commands or questions without providing examples. It leverages the LM pre-trained knowledge to solve problems and can be effectively used for various tasks without additional data [33].

2.2.2 Role Prompting. A technique that assigns a specific role to the LM, guiding it to interpret problems and generate responses from that perspective [32]. Also known as persona prompting [27], this approach enhances role-specific knowledge to produce more satisfactory answers.

2.2.3 (Zero-Shot) Chain of Thought Prompting. A technique that guides the LM to solve problems step by step, based on a Zero-Shot approach that leverages its pre-trained knowledge without providing examples. This method includes prompts like "Let's think

step by step" to generate logical reasoning and gradually resolve complex problems [13].

2.2.4 Self-Refine Prompting. A technique in which the LM provides feedback on its initial response and iteratively improves it. By utilizing the LM's self-evaluation capability, this method refines the initial output, generating more detailed and sophisticated results through multiple iterations [21].

2.2.5 Least-to-Most Prompting. A technique that breaks down a problem into simpler sub-problems, solving them sequentially to derive the final result. By guiding the LM to address smaller units of the problem one at a time, this approach enhances the overall logical coherence and accuracy of the task [36].

3 Methodology

Our study conducted a controlled experiment using the application of Musinsa, one of the largest shopping platforms in South Korea [12], to explore the most effective prompt engineering techniques for UX evaluation leveraging LLM. The shopping domain was selected because it is one of the fields where UX evaluation is most actively applied and includes complex user interface elements, such as product search, filtering, pagination, and advertisements, making it a suitable environment to explore the potential of LLM-based UX evaluation [25, 29, 35].

3.1 Procedure

Our study consisted of four main stages: (1) We analyzed the Musinsa application and designed four anticipated user scenarios (i.e., product search, viewing product details, adding to cart, and making a purchase) that reflect key features and user flows. Each scenario included representative tasks users could perform on the app and the interface elements they might encounter. (2) Screenshots of the application screens included in each scenario were collected for detailed analysis. (3) Input prompts were created based on five prompt engineering techniques. These prompts were

designed to help the LLM identify potential usability issues based on the interfaces in each scenario. (4) The final list of usability issues identified by the LLM was documented and organized. The dataset included each scenario, screen images, LLM-generated feedback, identified usability issues, and their classification categories.

3.2 Experiment Setup

3.2.1 Prior Study for LLM Model Selection. This study conducted a preliminary beta test using the Zero-Shot Prompting technique on ChatGPT-4 and Gemini 2.0, both of which support multimodal image processing, to select the experimental model [1, 30]. The results showed that ChatGPT-4 outperformed Gemini in terms of accuracy and analytical capability when processing text and image data. Gemini often exhibited hallucination phenomena, making it difficult to ensure the reliability of the results. Additionally, multiple prior studies have reported that ChatGPT-4 generally outperforms Gemini [10, 19]. For these reasons, ChatGPT-4 was selected as the final experimental model for this study.

3.2.2 Designing Comparative Prompts. The input prompts for ChatGPT-4 were designed based on the prepared scenarios and visual materials, following five prompting engineering techniques [13, 21, 32, 33, 36]. These prompts included content that aimed to clearly and specifically identify usability issues and accurately output the textual elements or UI components causing those issues (Figure 2). This design was intended to improve the accuracy and reliability of the generated responses, reduce hallucinations, and offer a foundation for researchers to objectively validate the responses.

We collected the textual responses generated by ChatGPT-4, which processed the input data based on the designed prompts. The responses were organized into a predefined data format for subsequent analysis and comparison of results. To ensure consistency across experiments, the same environment and process were applied to all techniques.

3.3 Data Analysis

The analysis of the generated responses by ChatGPT-4 evaluated how many UX issues were identified and assessed them based on Jakob Nielsen’s severity rating criteria [23]. Nielsen’s criteria are widely used to evaluate usability issues and prioritize their resolution. Following this framework, severity was evaluated on a 4-point scale (ranging from 0 to 4) according to three factors: frequency, impact, and persistence.

We qualitatively analyzed the UX issues identified and improvement suggestions generated by ChatGPT-4 based on these evaluation criteria. Two researchers independently performed the severity classification to ensure consistency, considering each issue’s characteristics and contextual background. After analyzing the same dataset individually, the researchers compared their results and reached a consensus to determine the final severity scores. Additionally, responses rated as 0 under the severity evaluation criteria were further classified into two categories: items not considered usability issues and error responses. Items not considered usability issues included cases where the identified issues did not impact usability, while error responses referred to outputs that provided irrelevant or contextually inappropriate answers. This additional

analysis was utilized to identify the limitations and abnormal response patterns of the LLM and to compare the frequency of errors based on prompting techniques.

4 Comparative Results of Prompt Engineering Techniques

The experimental results showed varying effectiveness among the five prompt engineering techniques (Zero-Shot, Role, Zero-Shot CoT, Self-Refining, Least-to-Most) in effectively identifying and analyzing UX issues within the context of shopping applications (Table 1). Zero-Shot Prompting identified a relatively small number of issues (Total identified issues = 15, Average severity score = 1.2), with responses often vague or general. In contrast, Role Prompting was more effective in accurately identifying issues, significantly improving the quality and relevance of results (Total identified issues = 20, Average severity score = 2.4). This technique frequently identified influential issues (Major usability problems, Score = 3), such as confusing terminology and complex navigation. In the product detail view scenario, it highlighted issues such as “Users may find terms like ‘catwashing’ or ‘long sleeve’ unintuitive, causing confusion,” utilizing specific persona perspectives to provide contextual analysis and actionable insights. The (Zero-Shot) CoT Prompting effectively identifies complex multi-step UX issues. Also, it uncovered more severe issues (Usability catastrophe, Score = 4) compared to other techniques (Total identified issues = 22, Average severity score = 2.5). In the product checkout scenario, it emphasized the issue of “There is no UI to return to the product page after canceling a payment,” showcasing its effectiveness in systematically analyzing complex interaction flows through step-by-step reasoning. The Self-Refining Prompting guided the model to iteratively review and refine its responses, producing more sophisticated analytical results (Total identified issues = 16, Average severity score = 2.1). The Least-to-Most Prompting technique comprehensively identified a wide range of issues, effectively detecting problems related to visual clarity and accessibility (Total identified issues = 19, Average severity score = 2.1).

Table 1: Total Number of Identified UX Issues and Average Severity Scores Across the Five Prompt Engineering Techniques: Based on Jakob Nielsen’s Severity Evaluation Criteria (0 = Not a problem; 1 = Cosmetic problem only; 2 = Minor usability problem; 3 = Major usability problem; 4 = Usability catastrophe).

Prompt Engineering Technique	Total identified UX Issues	Average Severity Score
Zero-Shot Prompting	15	1.2
Role Prompting	20	2.4
(Zero-Shot) Chain of Thought Prompting	22	2.5
Self-Refining Prompting	16	2.1
Least-to-Most Prompting	19	2.1

Ultimately, the Role Prompting and (Zero-Shot) CoT Prompting techniques proved to be the most effective in identifying UX issues. Specifically, the Role Prompting technique provided contextual

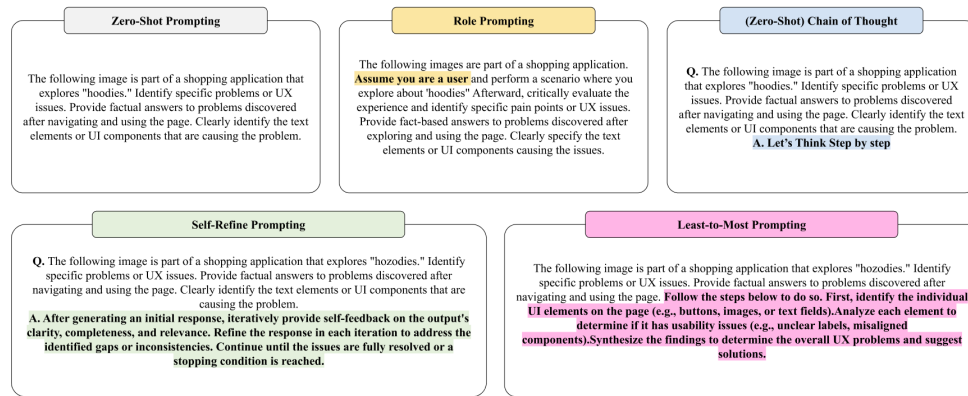


Figure 2: Comparison of Different Prompt Engineering Techniques for UX Evaluation: Examining various approaches, including Zero-Shot, Role, (Zero-Shot) CoT, Self-Refining, and Least-to-Most prompting techniques to identify and address UX issues within a shopping application interface.

analysis and actionable insights from a user perspective, while the CoT Prompting technique showed outstanding performance in systematically analyzing complex multi-step issues.

5 Hybrid Approach: Role Prompting + (Zero-Shot) Chain of Thought Prompting

After comparing the five prompt engineering techniques, we conducted an additional validation experiment by combining the two most effective techniques: Role Prompting and (Zero-Shot) CoT Prompting (Figure 3). This study was carried out to address the challenges of relying on a single technique and to explore a new approach for more effectively identifying UX issues.

The analysis revealed that the combination of Role Prompting and (Zero-Shot) CoT Prompting was significantly more effective than using each technique individually. In particular, the detailed contextual understanding provided by the Role Prompting technique and the systematic logical structure of (Zero-Shot) CoT Prompting created a synergistic effect in identifying UX issues.

This hybrid technique proved to be more effective in identifying high-severity issues, such as serious navigation errors and complex interaction problems spanning multiple screens. A total of 27 valid UX issues were identified, with an average severity score of 2.9. This demonstrates superior effectiveness compared to individual techniques, where Role Prompting identified 20 issues with an average severity score of 2.4, and (Zero-Shot) CoT Prompting identified 22 issues with an average severity score of 2.5. This indicates that the hybrid approach compensates for the insufficiency of individual techniques, such as the tendency of Role Prompting to generalize based on roles and the verbosity of responses caused by the explicit reasoning process of (Zero-Shot) CoT Prompting. It enables the discovery of more critical usability issues while generating clear and consistent responses.

However, other combinations were less effective in identifying UX issues compared to the Role + (Zero-Shot) CoT Hybrid approach. The combination of Role Prompting and Self-Refining Prompting was effective in detecting issues related to content clarity and visual hierarchy due to its iterative feedback structure. Still, it struggled

with workflows requiring multi-step reasoning, as it lacked the systematic logic provided by (Zero-Shot) CoT. Similarly, the combination of Role Prompting and Least-to-Most Prompting was useful for systematically identifying lower-priority issues but struggled to thoroughly uncover major UX issues.

Therefore, the Role + (Zero-Shot) CoT Hybrid approach emerged as the most effective combination, offering both systematic reasoning and contextual depth to identify a broader range of UX issues comprehensively.

6 Findings & Discussion

6.1 Patterns in LLM-Generated Responses

We qualitatively analyzed generated responses for UX evaluation by LLM, carefully examining the perspectives and details presented in each response. The outputs revealed unique patterns, including insights that can be overlooked in traditional UX evaluation processes or conclusions that differ from those of human evaluators.

6.1.1 Identify Tiny and Detailed Issues That Humans Might Overlook. The UX evaluation responses generated by the LLM revealed a distinctive ability to identify tiny and detailed UX issues often overlooked by human evaluators (e.g., subtle text sizes, slightly misaligned layouts) while also proposing fresh perspectives. During the analysis of a product review interface, the LLM pointed out that excessively long reviews might cause users to miss critical information and suggested implementing a review summarization feature. Additionally, it provided specific feedback such as, "The text size of reviews is too small, reducing readability. Furthermore, the lack of clear distinction between positive and negative reviews makes it difficult for users to filter relevant information." These insights, often missed by human UX evaluators, highlight the LLM's strength in analyzing large volumes of data in detail to identify UX issues. This is also consistent with previous studies that demonstrated LLM's ability to effectively identify subtle issues that are often overlooked [17, 20]. It shows the potential of LLM to effectively uncover usability problems that human evaluators might overlook by processing and analyzing large volumes of text and UI components.

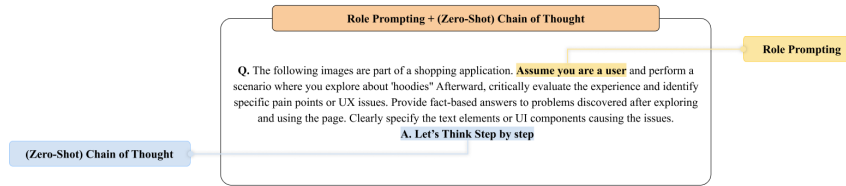


Figure 3: Hybrid Prompting Approach for UX Evaluation: Role Prompting + (Zero-Shot) Chain of Thought.

6.1.2 Hyper-accuracy Distortion in LLM-Generated Responses. One of the notable findings observed in the LLM's responses is a phenomenon stemming from the gap between human visual experience and the model's logical analysis, which we term 'Hyper-accuracy distortion [2]'. This highlights how LLM fails to replicate human intuitive judgments, instead repeatedly generating mechanical and logical evaluations. ChatGPT-4 assessed the 'Checkout' action button, which human users intuitively judged as visually clear, as having "Low visibility due to the frequent repetition of the same color tone and its lack of central placement on the screen." (Figure 4)

This case illustrates that LLM fails to grasp the visual compensation and contextual judgments humans unconsciously perform [16]. Even when a button is slightly off-center or features repetitive color tones, humans consider the overall context and surrounding design elements to judge the button as sufficiently clear and functional intuitively. In contrast, LLM, lacking this contextual understanding, mechanically analyze factors like color contrast and spatial placement, leading to overly critical evaluations. This can cause LLM to excessively analyze designs that are considered good enough from a human experiential perspective, raising unnecessary problems.

6.2 Challenges to Address in LLM-Based UX Evaluation

To address the challenges of LLM in effectively identifying and evaluating UX issues, a new approach that harmonizes human-centered design principles with mechanical analysis is needed. By achieving this balance, it is possible to enhance the accuracy of UX evaluations and maximize the effectiveness of LLMs as UX evaluators.

6.2.1 Expanding Training Data for LLM to Learn Human Visual Compensation. Additional training datasets incorporating perceptual data are essential to enable LLM to better understand the human visual compensation process. By simulating how humans visually process UI elements or incorporating visual context data into model training, LLM could address existing limitations and enhance their ability to evaluate UX. This approach would allow LLM to complement mechanical analysis with a more user-centered perspective, effectively supporting UX evaluations by considering contextual and perceptual nuances.

6.2.2 Incorporate Human-Centered Perspectives into LLM Prompt Design. When designing prompts for LLM-based UX evaluations, it is essential to utilize prompt engineering techniques that incorporate human-centered perceptual elements rather than merely requesting logical analysis. Prompts that include questions like,

"How might this button appear to the user?" can guide the LLM to emulate human visual experiences more closely. Such prompting techniques enable LLM to provide analyses that are more aligned with human perspectives during UX evaluations.

These approaches play an important role in bridging the gap between LLM and human evaluations, thereby enhancing the effectiveness and precision of UX evaluation. A new evaluation paradigm based on the complementary collaboration between humans and LLM can offer more profound and integrated insights.

7 Limitation & Future Work

This section discusses the study's limitations and proposes directions for future research. First, we utilized ChatGPT-4, one of the most advanced and widely used models. While this allowed us to apply cutting-edge technology to UX evaluation, it is possible that the capabilities and limitations of this specific model influenced the findings to some extent. ChatGPT-4 is a proprietary model developed and released by OpenAI, meaning researchers had no direct control over its architecture and inherent characteristics. Future studies could expand the scope of research by developing custom LLM tailored for UX evaluation or employing a variety of LLM, including the latest versions.

Second, this study relied on static images and text-based analysis, which may have excluded real-time user interactions and dynamic usage behaviors. UX evaluation often involves dynamic interactions and real-time observation. However, ChatGPT-4 does not currently support real-time interaction capabilities. Therefore, future research should consider integrating dynamic interaction simulations or exploring multimodal LLM capable of processing real-time user behavior and feedback.

8 Conclusion

In this study, we explored the most effective prompt engineering techniques for identifying UX issues using a multimodal LLM. To achieve this, we conducted a comparative analysis of five prompt engineering techniques using ChatGPT-4. As a result, Role Prompting and (Zero-Shot) CoT Prompting were found to be particularly effective in identifying and analyzing usability issues by employing user-centered contextual analysis and systematic problem-solving approaches. Notably, the hybrid approach combining these two techniques maximized their respective strengths and showed the highest effectiveness in identifying critical UX issues. Moreover, hyper-accuracy distortion observed during the LLM's analysis process presented new perspectives distinct from human intuitive

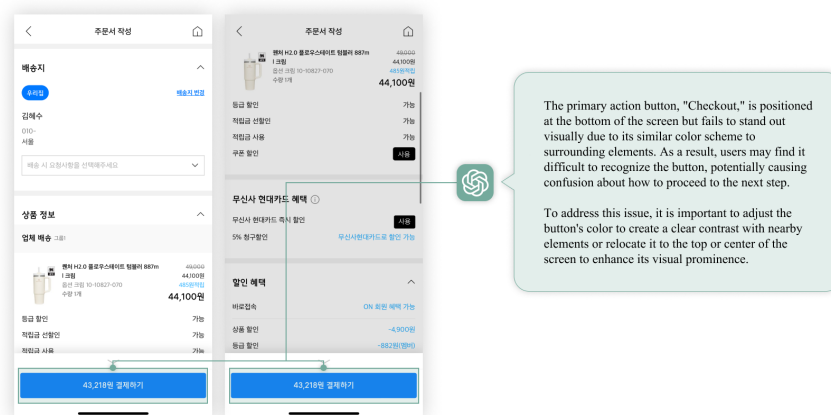


Figure 4: ChatGPT-4's Evaluation of the Visual Visibility of the Blue 'Checkout' Button Located at the Bottom of the Interface.

judgment, opening up the possibility of discovering specific considerations for prompt design in UX evaluation, such as mechanical analysis standards for visual elements, the potential for errors due to the lack of contextual judgment, and the redefinition of acceptable imperfections. These findings suggest the practical possibility of multimodal LLM-based UX evaluation tools. They are expected to be effectively and usefully applied in future research to explore the intrinsic needs of more diverse user groups.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).
- [2] Gati V Aher, Rosa I Arriaga, and Adam Tauman Kalai. 2023. Using large language models to simulate multiple humans and replicate human subject studies. In *International Conference on Machine Learning*. PMLR, 337–371.
- [3] Xavier Amatriain. 2024. Prompt design and engineering: Introduction and advanced methods. *arXiv preprint arXiv:2401.14423* (2024).
- [4] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.
- [5] Banghao Chen, Zhaofeng Zhang, Nicolas Langrené, and Shengxin Zhu. 2023. Unleashing the potential of prompt engineering in Large Language Models: a comprehensive review. *arXiv preprint arXiv:2310.14735* (2023).
- [6] Mingming Fan, Serina Shi, and Khai N Truong. 2020. Practices and Challenges of Using Think-Aloud Protocols in Industry: An International Survey. *Journal of Usability Studies* 15, 2 (2020).
- [7] Mingming Fan, Xianyou Yang, TszTung Yu, Q Vera Liao, and Jian Zhao. 2022. Human-ai collaboration for UX evaluation: effects of explanation and synchronization. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW1 (2022), 1–32.
- [8] Sergio Firmenich, Alejandra Garrido, Julián Grigera, José Matías Rivero, and Gustavo Rossi. 2019. Usability improvement through A/B testing and refactoring. *Software Quality Journal* 27 (2019), 203–240.
- [9] Asbjørn Følstad, Effie Law, and Kasper Hornbæk. 2012. Analysis in practical usability evaluation: a survey study. In *proceedings of the SIGCHI conference on human factors in computing systems*. 2127–2136.
- [10] Gurnoor S Gill, Joby Tsai, Jillene Moxam, Harshal A Sanghvi, and Shailesh Gupta. 2024. Comparison of Gemini Advanced and ChatGPT 4.0's Performances on the Ophthalmology Resident Ophthalmic Knowledge Assessment Program (OKAP) Examination Review Question Banks. *Cureus* 16, 9 (2024), e69612.
- [11] Louie Giray. 2023. Prompt engineering with ChatGPT: a guide for academic writers. *Annals of biomedical engineering* 51, 12 (2023), 2629–2633.
- [12] Lee Hyo-sik. 2023. Musinsa joins list of world's most innovative companies for 2023. https://www.koreatimes.co.kr/www/tech/2025/01/129_346457.html. Accessed: 22 January 2025.
- [13] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. *Advances in neural information processing systems* 35 (2022), 22199–22213.
- [14] Emily Kuang, Minghao Li, Mingming Fan, and Kristen Shinohara. 2024. Enhancing UX Evaluation Through Collaboration with Conversational AI Assistants: Effects of Proactive Dialogue and Timing. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–16.
- [15] Gyeong-Geon Lee, Ehsan Latif, Lehong Shi, and Xiaoming Zhai. 2023. Gemini pro defeated by gpt-4v: Evidence from education. *arXiv preprint arXiv:2401.08660* (2023).
- [16] Steven M Lehar. 2003. *The world in your head: A gestalt view of the mechanism of conscious experience*. Psychology Press.
- [17] Ziyu Li and Donghwan Shin. 2024. Mutation-based consistency testing for evaluating the code understanding capability of llms. In *Proceedings of the IEEE/ACM 3rd International Conference on AI Engineering-Software Engineering for AI*. 150–159.
- [18] Mengchen Liu, Chongyan Chen, and Danna Gurari. 2023. An evaluation of gpt-4v and gemini in online vqa. *arXiv preprint arXiv:2312.10637* (2023).
- [19] Mengchen Liu, Chongyan Chen, and Danna Gurari. 2023. An evaluation of gpt-4v and gemini in online vqa. *arXiv preprint arXiv:2312.10637* (2023).
- [20] Stephen MacNeil, Paul Denny, Andrew Tran, Juho Leinonen, Seth Bernstein, Arto Hellas, Sami Sarsa, and Joanne Kim. 2024. Decoding logic errors: a comparative study on bug detection by students and large language models. In *Proceedings of the 26th Australasian Computing Education Conference*. 11–18.
- [21] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. 2024. Self-refine: Iterative refinement with self-feedback. *Advances in Neural Information Processing Systems* 36 (2024).
- [22] Bertalan Meskó. 2023. Prompt engineering as an important emerging skill for medical professionals: tutorial. *Journal of medical Internet research* 25 (2023), e50638.
- [23] Jakob Nielsen. 1994. Usability inspection methods. In *Conference companion on Human factors in computing systems*. 413–414.
- [24] Mie Nørgaard and Kasper Hornbæk. 2006. What do usability evaluators do in practice? An explorative study of think-aloud testing. In *Proceedings of the 6th conference on Designing Interactive systems*. 209–218.
- [25] Ilias O Pappas. 2018. User experience in personalized online shopping: a fuzzy-set analysis. *European Journal of Marketing* 52, 7/8 (2018), 1679–1703.
- [26] Daeseung Park, Gi-taek An, Chayapol Kamyod, and Cheong Ghil Kim. 2023. A Study on Performance Improvement of Prompt Engineering for Generative AI with a Large Language Model. *Journal of Web Engineering* 22, 8 (2023), 1187–1206.
- [27] Douglas C Schmidt, Jesse Spencer-Smith, Quchen Fu, and Jules White. 2023. Cataloging prompt patterns to enhance the discipline of prompt engineering. URL: https://www.dre.vanderbilt.edu/~schmidt/PDF/ADA_Europe_Position_Paper.pdf [accessed 2023-09-25] (2023).
- [28] Sander Schulhoff, Michael Ilie, Nishant Balepur, Konstantine Kahadze, Amanda Liu, Chenglei Si, Yinheng Li, Aayush Gupta, Hyojung Han, Sevien Schulhoff, et al. 2024. The Prompt Report: A Systematic Survey of Prompting Techniques. *arXiv preprint arXiv:2406.06608* (2024).
- [29] Stefanie Sohn, Barbara Seegerbarth, and Madleen Moritz. 2017. The impact of perceived visual complexity of mobile online shops on user's satisfaction. *Psychology & Marketing* 34, 2 (2017), 195–214.

- [30] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. 2023. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805* (2023).
- [31] Jiaqi Wang, Zhengliang Liu, Lin Zhao, Zihao Wu, Chong Ma, Sigang Yu, Haixing Dai, Qiushi Yang, Yiheng Liu, Songyao Zhang, et al. 2023. Review of large vision models and visual prompt engineering. *Meta-Radiology* (2023), 100047.
- [32] Zekun Moore Wang, Zhongyuan Peng, Haoran Que, Jiaheng Liu, Wangchunshu Zhou, Yuhan Wu, Hongcheng Guo, Ruitong Gan, Zehao Ni, Jian Yang, et al. 2023. Rolellm: Benchmarking, eliciting, and enhancing role-playing abilities of large language models. *arXiv preprint arXiv:2310.00746* (2023).
- [33] Jason Wei, Maarten Bosma, Vincent Y Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M Dai, and Quoc V Le. 2021. Finetuned language models are zero-shot learners. *arXiv preprint arXiv:2109.01652* (2021).
- [34] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C Schmidt. 2023. A prompt pattern catalog to enhance prompt engineering with chatgpt. *arXiv preprint arXiv:2302.11382* (2023).
- [35] Ling Zheng and Long Xu. 2020. A Study on Evaluation Model of VR Shopping Experience based on User-Centered Approach. In *2020 International Conference on Innovation Design and Digital Technology (ICIDDT)*. IEEE, 323–333.
- [36] Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc Le, et al. 2022. Least-to-most prompting enables complex reasoning in large language models. *arXiv preprint arXiv:2205.10625* (2022).